# Harsh Kumar

*Postdoctoral Fellow, Center for Astrophysics | Harvard & Smithsonian*
*Junior Investigator, NSF AI Institute IAIFI*

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## Academic Positions

2023–now **Postdoctoral Fellow in Astrophysics**, Center for Astrophysics | Harvard & Smithsonian (advisor: Prof. Edo Berger); **Junior Investigator**, NSF AI Institute for Artificial Intelligence and Fundamental Interactions (IAIFI).

2023 **Institute Postdoctoral Fellow**, Indian Institute of Technology Bombay, Mumbai (May–August).

2023 **Research Associate**, Indian Institute of Technology Bombay, Mumbai (April–May).

## Education

2018–2023 **Ph.D. in Astrophysics**, IIT Bombay (advisor: Prof. Varun Bhalerao; course CGPA 9.3/10). Thesis: "Studying optical counterparts of gravitational waves and gamma-ray burst afterglows with the GROWTH-India Telescope."

2012–2017 **M.Sc. Astrophysics and Nuclear Physics** (2017, 69.6%) and **B.Sc. (Hons.) Physics** (2015, 90.3%), University of Delhi.

## Research Highlights

- **SN 2024afav** (first author, ApJL 2026): ties the post-peak light-curve bumps of an SLSN-I to discrete episodes of circumstellar interaction, using photometry from five sources and spectra from eight instruments on a single object.
- **Helium in SLSN-I progenitors** (first author, ApJ 2025a, b): a near-infrared detection of helium in the bright SN 2024rmj and an upper limit in SN 2024ahr, bracketing the helium an SLSN-I progenitor can retain.
- **Bump census** (single author, in preparation): after correction for completeness and false positives, 63% (68% interval 56–69%) of SLSNe-I have at least one light-curve bump of 0.2–0.8 mag.

## Research Funding (NASA, through STScI; Co-Investigator)

2026 HST-GO-18392, Cycle 34, PI X. Guo, 56 orbits — **$176,256**. *Probing SMBH Locations, Properties, and Accretion Physics with Late-time Observations of TDEs.*

2024–26 HST-GO-17738/17991/18449, Cycles 32–34, PI P. Blanchard, 25 orbits — **$171,795**. *Probing Magnetar Energy Input with the First Sample of Late-Time Superluminous Supernovae.*

2026 JWST-GO-11372, Cycle 5, PI P. Blanchard, 6.9 h — **$110,478**. *Is the NIR Counterpart of FRB 20250316A a Transient?*

2025 JWST-DD-12468, Cycle 4, PI D. Hiramatsu, 4.6 h — **$35,000**. *The First Near-Infrared Nebular Spectrum of a Pair-Instability Supernova.*

Total **$493,529** (program allocations in the STScI budget notifications, all investigators combined).

## Telescope Time

As PI **40 programs** allocated, 2024A–2026B: Gemini North and South 15 (one Director's Discretionary), MMT 13, FLWO 1.2 m 9, Magellan 3 (of 30 submitted to MMT, Magellan, and FLWO, 25 were allocated); three more under review for 2027A.

PI or Co-I Gemini North 9 and Gemini South 11 accepted proposals since 2023; MMT ≥20 (2024A–2026A); a 3-year SOAR/Goodman program (Co-I); Himalayan Chandra Telescope 9 (2019–2022).

## Publications

**52 refereed publications**, **7 as first author**; co-author of 3 papers in *Nature* and 2 in *Nature Astronomy*; >200 GCN, ATel, and TNS reports and contributions to >100 Minor Planet Electronic Circulars. Full list: NASA ADS / ORCID.

**First author:**

1. **Kumar, H.**, et al. 2026, ApJL, 998, L3 — SN 2024afav: light-curve bumps tied to circumstellar interaction.
2. **Kumar, H.**, et al. 2025, ApJ, 992, 122 — a detection of helium in the bright SLSN-I SN 2024rmj.
3. **Kumar, H.**, et al. 2025, ApJ, 987, 127 — a near-infrared search for helium in SN 2024ahr.
4. **Kumar, H.**, et al. 2024, ApJL, 974, L36 — AT 2023vto, an exceptionally luminous helium TDE from a massive star.
5. **Kumar, H.**, et al. 2022, AJ, 164, 90 — the GROWTH-India Telescope, India's first robotic telescope.
6. **Kumar, H.**, et al. 2022, MNRAS, 516, 4517 — the GROWTH-India search for S190426c with a custom vetting pipeline.
7. **Kumar, H.**, et al. 2022, MNRAS, 513, 2777 — GRB 210204A, the most delayed flares in a gamma-ray-burst afterglow.

**Selected co-authored:**

8. Farah, J. R., et al. (+ **Kumar, H.**) 2026, *Nature*, 651, 321 — a precessing magnetar engine in an SLSN-I.
9. Ho, A. Y. Q., et al. (+ **Kumar, H.**) 2023, *Nature*, 623, 927 — minutes-duration optical flares; I planned the HCT and GIT observations simultaneous with Chandra and led their reduction and analysis.
10. Andreoni, I., et al. (+ **Kumar, H.**) 2022, *Nature*, 612, 430 — a luminous jetted TDE; I led the Indian follow-up (planning, acquisition, and reduction).

11. Ahumada, T., et al. (+ **Kumar, H.**) 2021, *Nature Astronomy*, 5, 917 — the shortest gamma-ray burst from a collapsar.
12. Anand, S., et al. (+ **Kumar, H.**) 2021, *Nature Astronomy*, 5, 46 — optical follow-up of two NS–BH mergers.
13. Swain, V., Bhalerao, V., **Kumar, H.**, et al. 2026, ApJ, 1000, 58 — GRB 230204B, a fast-fading afterglow found with GIT; first author co-mentored by me.
14. Sharma, K., **Kumar, H.**, et al. 2023, MNRAS, 524, 2651 — Astreaks, NEO astrometry from trailed stars; first author co-mentored by me.

**In preparation (single author):** "Bumps and Interaction Signatures in Hydrogen-poor Superluminous Supernovae" I (light-curve census) and II (spectroscopic census and its connection to the bumps).

## Teaching

- **Teaching assistant**, Department of Physics, IIT Bombay, for eight courses: Astrophysics I and II (PH426, PH556), Programming Lab (PH434), and five electronics laboratories (PH230, PH231, PH233, PH435/PH503, PH443); ran laboratory and tutorial sessions, designed assignments, graded examinations, and occasionally lectured. No numerical student-evaluation scores were issued for these roles.
- **Tutorials** at Transient Tea, CfA (2024–2026), on key topics in transient astronomy for students and junior researchers; **certified Carpentries instructor**; co-organizer, First GROWTH-India Winter School (2018); tutor, GROWTH Astronomy School (2019); mentor, Krittika summer projects, IIT Bombay.

## Mentoring and Supervision

**More than 25 students co-mentored** (2018–2026; Ph.D., M.S., and B.S. projects) through GROWTH-India, the CfA, and IAIFI. Eight went on to Ph.D. programs. Outcomes: H. Iyer (Ph.D., UIUC), V. Karambelkar (Ph.D., Caltech; NASA Hubble Fellow), K. Sharma (Ph.D., Caltech), S. Joharle (Ph.D., MPIA/Heidelberg), G. Waratkar (Ph.D., IIT Bombay; now at Caltech), A. Jasani (M.S., Toronto). Three co-mentees first-authored refereed papers (Swain and Sharma, listed above; T. Roychowdhury, JApA, 46, 68). At the CfA: K. Yadavalli (FLWO 1.2 m photometry and image-subtraction pipeline, 2024–25); A. de Pino (Harvard undergraduate, 2026); current: A. Joshi (master's, IISER Pune; pre-discovery mass ejection in SLSNe-I) and A. Bhatnagar (junior college; magnetar-model correlations).

## Service and Leadership

- **Review panels:** U.S. NSF panel reviewer (2024); time allocation committees for NOIRLab community time (Gemini/SOAR/NEID, 2024–2026), the 3.6-m Devasthal Optical Telescope (ARIES, 2021–2024), and the GMRT (NCRA–TIFR, 2023); referee for ApJ, ApJL, and PASP (2023–present).
- **Leadership:** GROWTH-India Telescope Development Team Leader, IIT Bombay / IIA (2018–2022); organizer and host, Transient Tea, CfA (2024–2026); SOC, Young Astronomers' Meet, ARIES (2022); coordinator, Astro–Cosmo–Gravity Journal Club, IIT Bombay (2018–2019); lifetime member, Astronomical Society of India.
- **Outreach:** school astronomy workshops in the Boston area (2024–present); the LIGO public exhibition at IIT Bombay's Techfest; Explore the Universe, Cambridge; National Science Day public lecture, IIT Bombay (2021).

## Honours and Awards

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| 2025 | **Zubin Kembhavi Award** (team award, GROWTH-India Telescope team), Astronomical Society of India. |
| 2023 | **Justice Oak Award for Outstanding Thesis in Astronomy**, Astronomical Society of India (ASI). |
| 2020–22 | Young Astronomer Award, Honourable Mention (ASI, 2022); Best Poster Award (ASI, 2021); New Discovery Award, Special Mention (ASI, 2020). |
| 2012–20 | **LSSTC Data Science Fellowship** (2018–2020); GATE Scholarship (2017); CSIR-NET Fellowship (2016); Central Sector Scholarship (2012–2017). |

## Invited Talks and Presentations

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| Invited | Colloquia at IIA, RRI, and IUCAA (2025–26); department seminars at IIT Bombay, IIT Hyderabad, and IISc (2025–26); IISER Mohali and IISER Pune (2025); ASI Best-Thesis Award lecture and ASI workshop on small-telescope networks (2024); Indian Physics Association, IIT Roorkee (2023); ZTF Summer School (2022). |
| Contributed | *Superluminous 2025* (Cambridge, USA); ESO Supernova Planetarium, Garching (2025); 39th and 41st ASI Meetings; XXXIst IAU General Assembly (Busan, e-talk). |

## Observing, Software, and Infrastructure

- **Facilities:** Gemini North and South (GMOS, GNIRS imaging and spectroscopy, image subtraction); MMT (Binospec, MMIRS); Magellan (IMACS, LDSS3, FIRE); SOAR (Goodman); FLWO 1.2 m (KeplerCam); HCT, 3.6-m DOT, and 1.3-m DFOT; >4 years of observing with the GROWTH-India Telescope.
- **Built:** automation, scheduling, and reduction pipelines of the GROWTH-India Telescope; a PypeIt-based multi-instrument reduction system (Binospec, Goodman, LDSS3; GNIRS, Flamingos-2, MMIRS with nodding); a real-time alert-classification, prioritization, and reduction backend serving the CfA time-domain program; the FLWO 1.2 m photometry and image-subtraction pipeline (with K. Yadavalli); photometry pipelines for the DOT and DFOT.
- **Methods and tools:** Python, C/C++, SQL; Astropy, PypeIt, ZOGY, MCMC (*emcee*, *dynesty*), Gaussian processes, Bayesian model comparison, injection-recovery completeness; broker APIs (TNS, Lasair, GCN), TOM Toolkit.

*Last updated: 23 October 2026.*